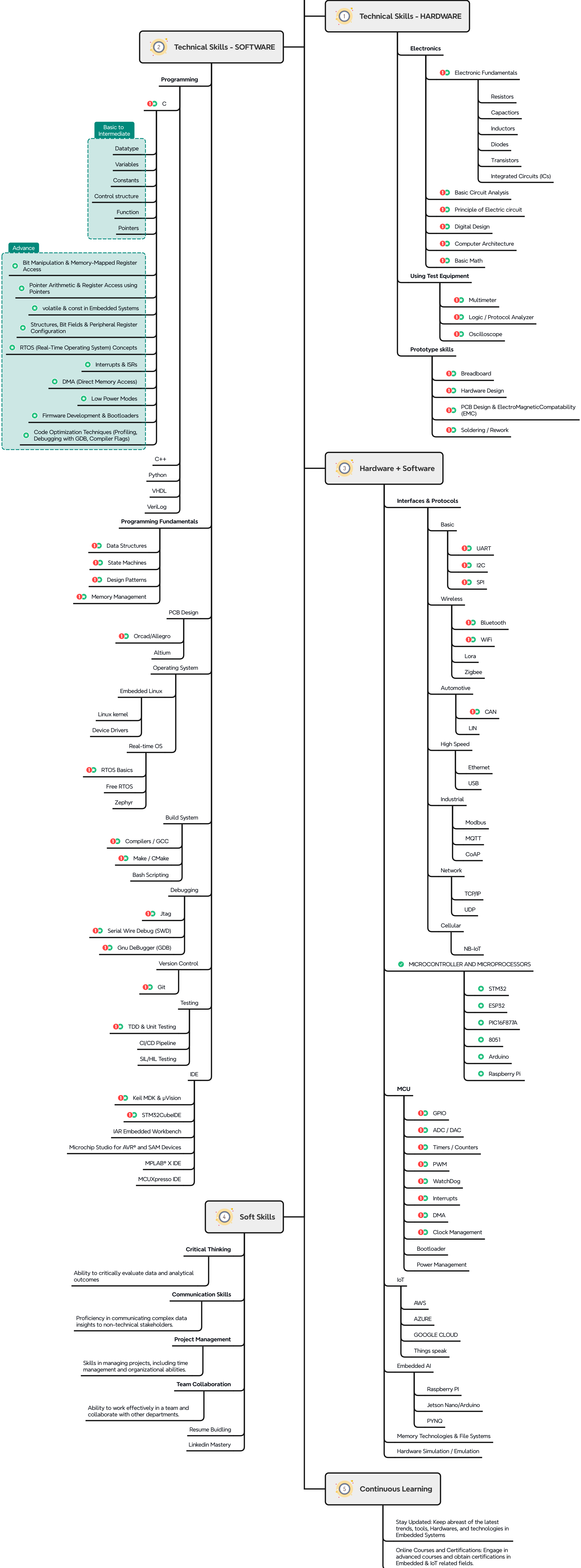




Road Map to become EMBEDDED SYSTEM ENGINEER



Programming

1

C

Basic to Intermediate

Datatype

Variables

Constants

Control structure

Function

Pointers

Advance

Bit Manipulation & Memory-Mapped Register Access

Pointer Arithmetic & Register Access using Pointers

volatile & const in Embedded Systems

Structures, Bit Fields & Peripheral Register Configuration

RTOS (Real-Time Operating System) Concepts

Interrupts & ISRs

DMA (Direct Memory Access)

Low Power Modes

Firmware Development & Bootloaders

Code Optimization Techniques (Profiling, Debugging with GDB, Compiler Flags)

C++

Python

VHDL

VeriLog

Programming Fundamentals

1

Data Structures

1

State Machines

1

Design Patterns

1

Memory Management

PCB Design

1

Orcad/Allegro

Altium

Operating System

Embedded Linux

Linux kernel

Device Drivers

Real-time OS

1

RTOS Basics

Free RTOS

Zephyr

Build System

1

Compilers / GCC

1

Make / CMake

Bash Scripting

Debugging

1

Jtag

1

Serial Wire Debug (SWD)

1

Gnu DeBugger (GDB)

Version Control

1

Git

Testing

1

TDD & Unit Testing

CI/CD Pipeline

SIL/HIL Testing

IDE

1

Keil MDK & uVision

1

STM32CubeIDE

IAR Embedded Workbench

Microchip Studio for AVR® and SAM Devices

MPLAB® X IDE

MCUXpresso IDE

Interfaces & Protocols

Basic

1

UART

1

I2C

1

SPI

Wireless

1

Bluetooth

1

WIFI

Lora

Zigbee

Automotive

1

CAN

LIN

High Speed

Ethernet

USB

Industrial

Modbus

MQTT

CoAP

Network

TCP/IP

UDP

Cellular

NB-IoT

MICROCONTROLLER AND MICROPROCESSORS

+

STM32

+

ESP32

+

PIC16F877A

+

8051

+

Arduino

+

Raspberry Pi

MCU

1

GPIO

1

ADC / DAC

1

Timers / Counters

1

PWM

1

WatchDog

1

Interrupts

1

DMA

1

Clock Management

Bootloader

Power Management

IoT

AWS

AZURE

GOOGLE CLOUD

Things speak

Embedded AI

Raspberry PI

Jetson Nano/Arduino

PYNQ

Memory Technologies & File Systems

Hardware Simulation / Emulation

Critical Thinking

Ability to critically evaluate data and analytical outcomes

Communication Skills

Proficiency in communicating complex data insights to non-technical stakeholders.

Project Management

Skills in managing projects, including time management and organizational abilities.

Team Collaboration

Ability to work effectively in a team and collaborate with other departments.

Resume Buidling

Linkedin Mastery

Stay Updated: Keep abreast of the latest trends, tools, Hardwares, and technologies in Embedded Systems

Online Courses and Certifications: Engage in advanced courses and obtain certifications in Embedded & IoT related fields.